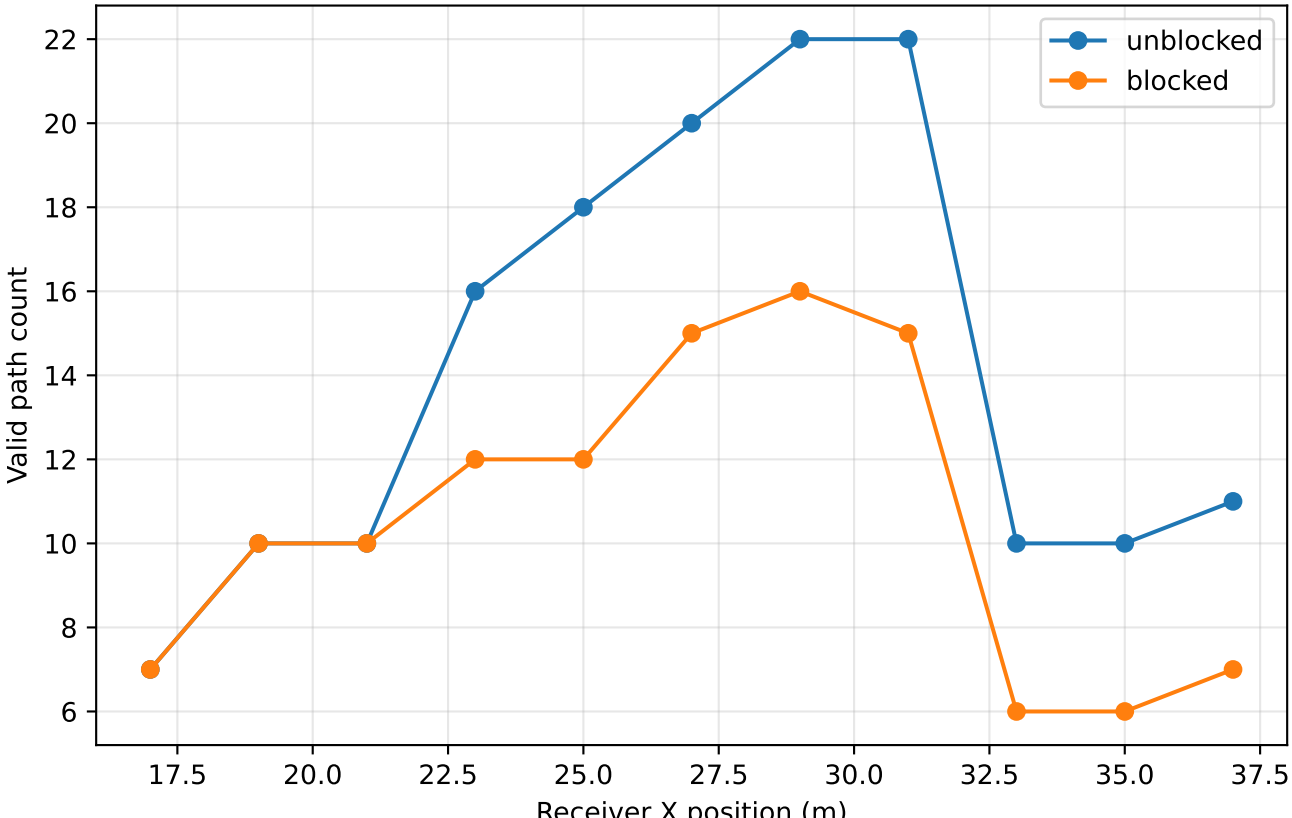
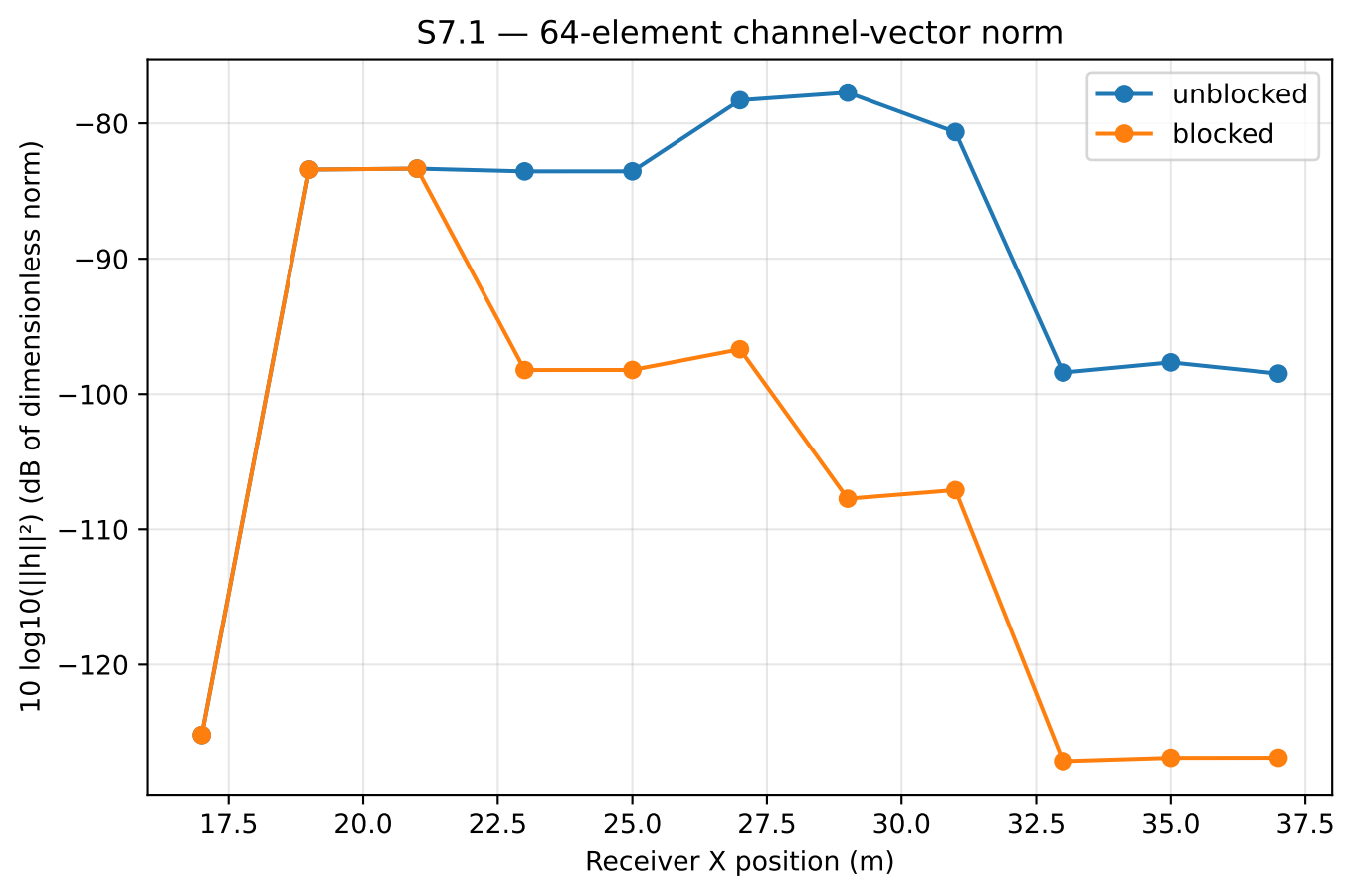
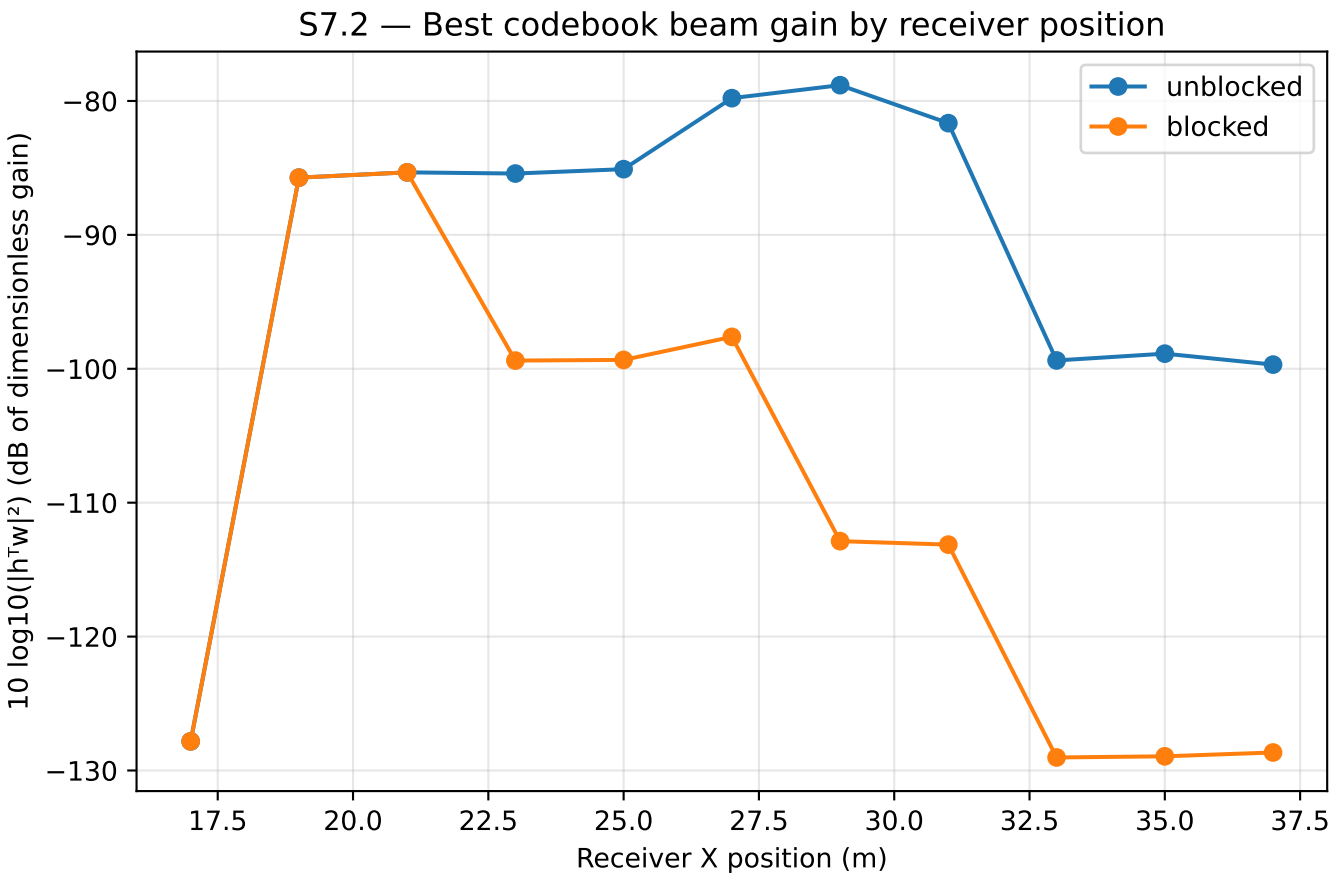
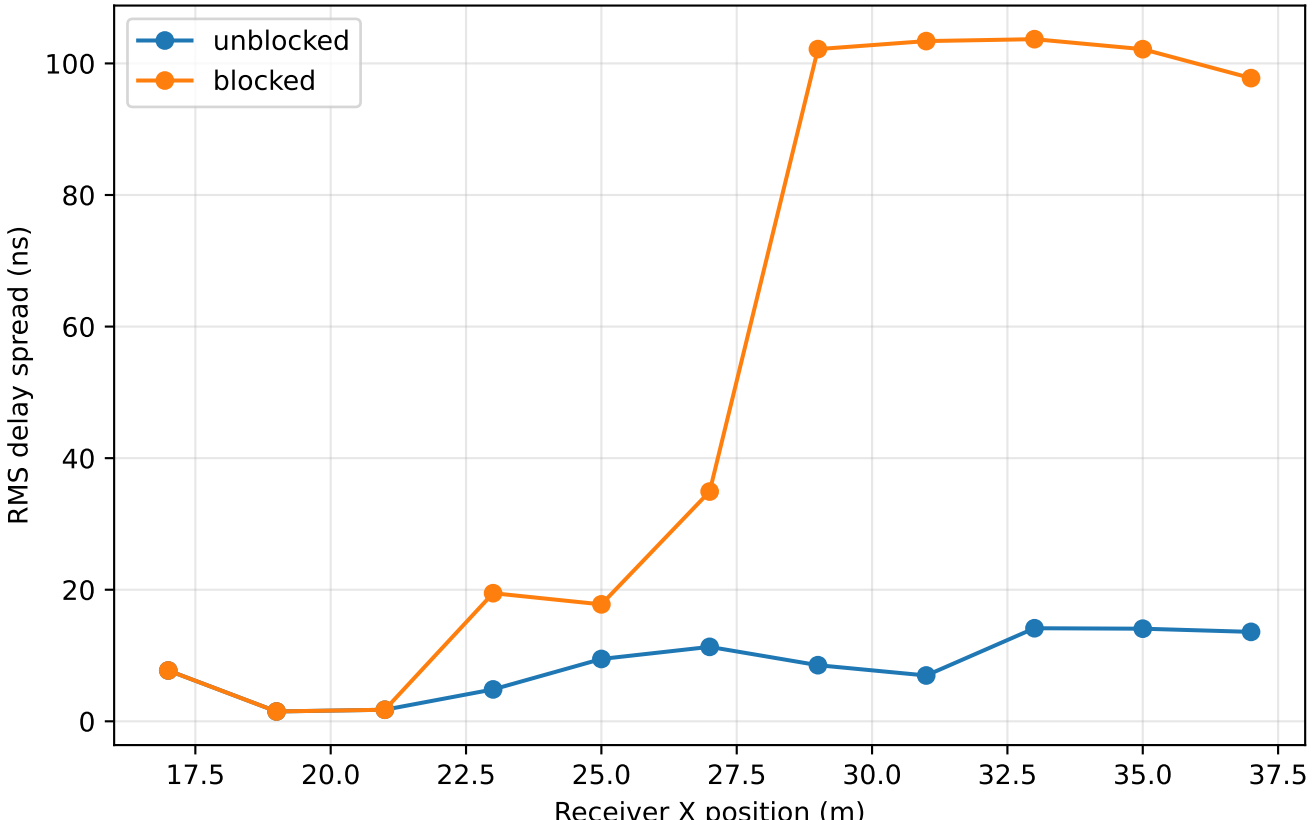
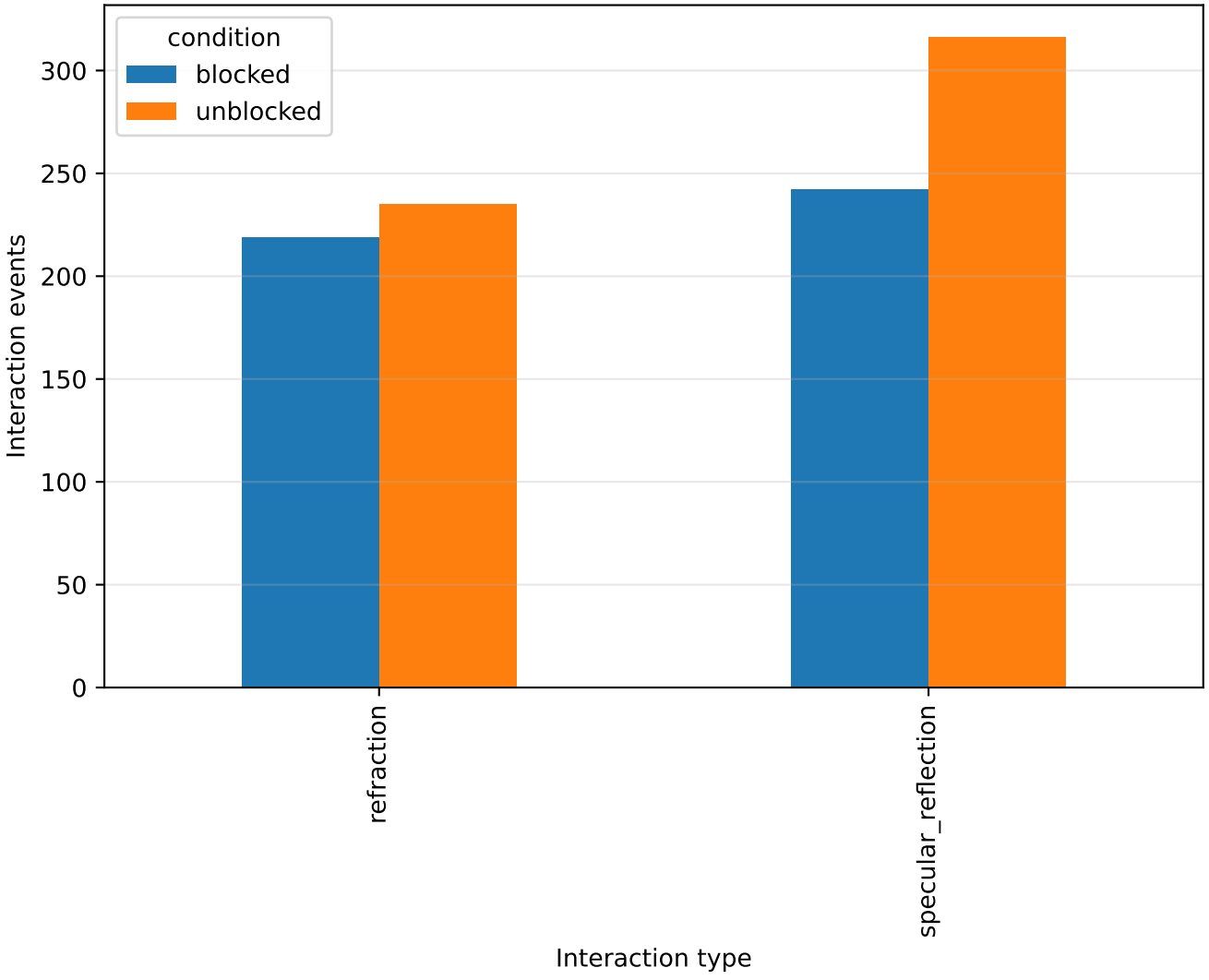
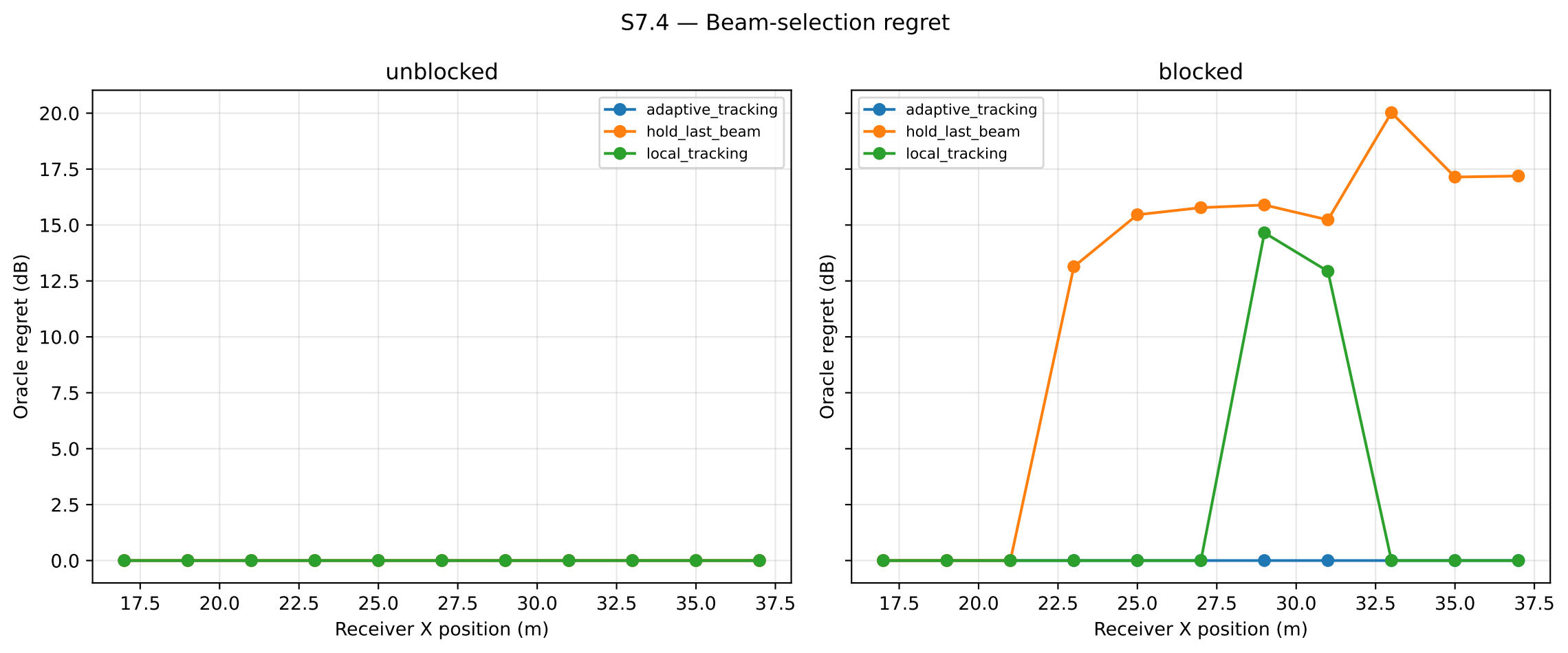

S7.1 — Valid ray-traced paths by receiver position
unblocked
blocked
Valid path count
Receiver X position (m)
22
20
18
16
14
12
10
8
6
17.5
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5

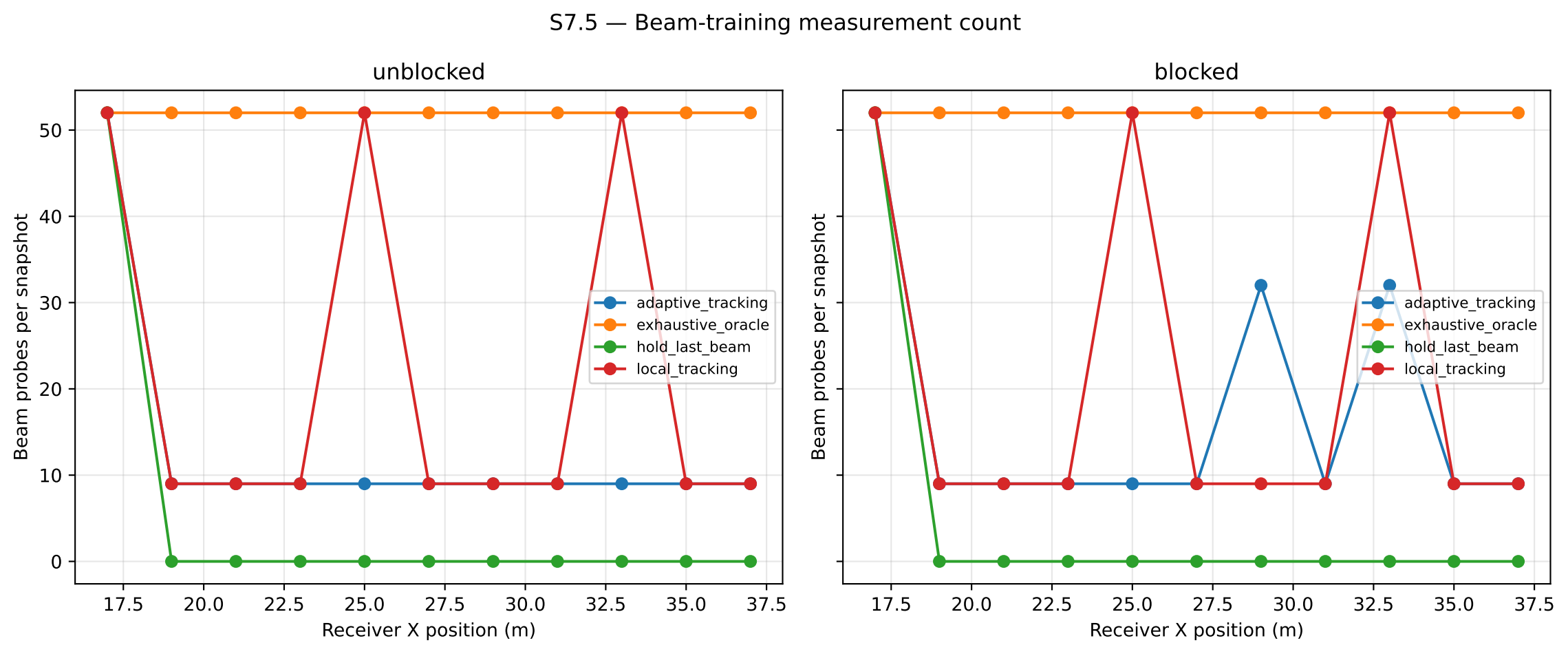

S7.1 — 64-element channel-vector norm
unblocked
blocked
−80
−90
−100
−110
−120
10 log10(||h||²) (dB of dimensionless norm)
17.5
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5
Receiver X position (m)

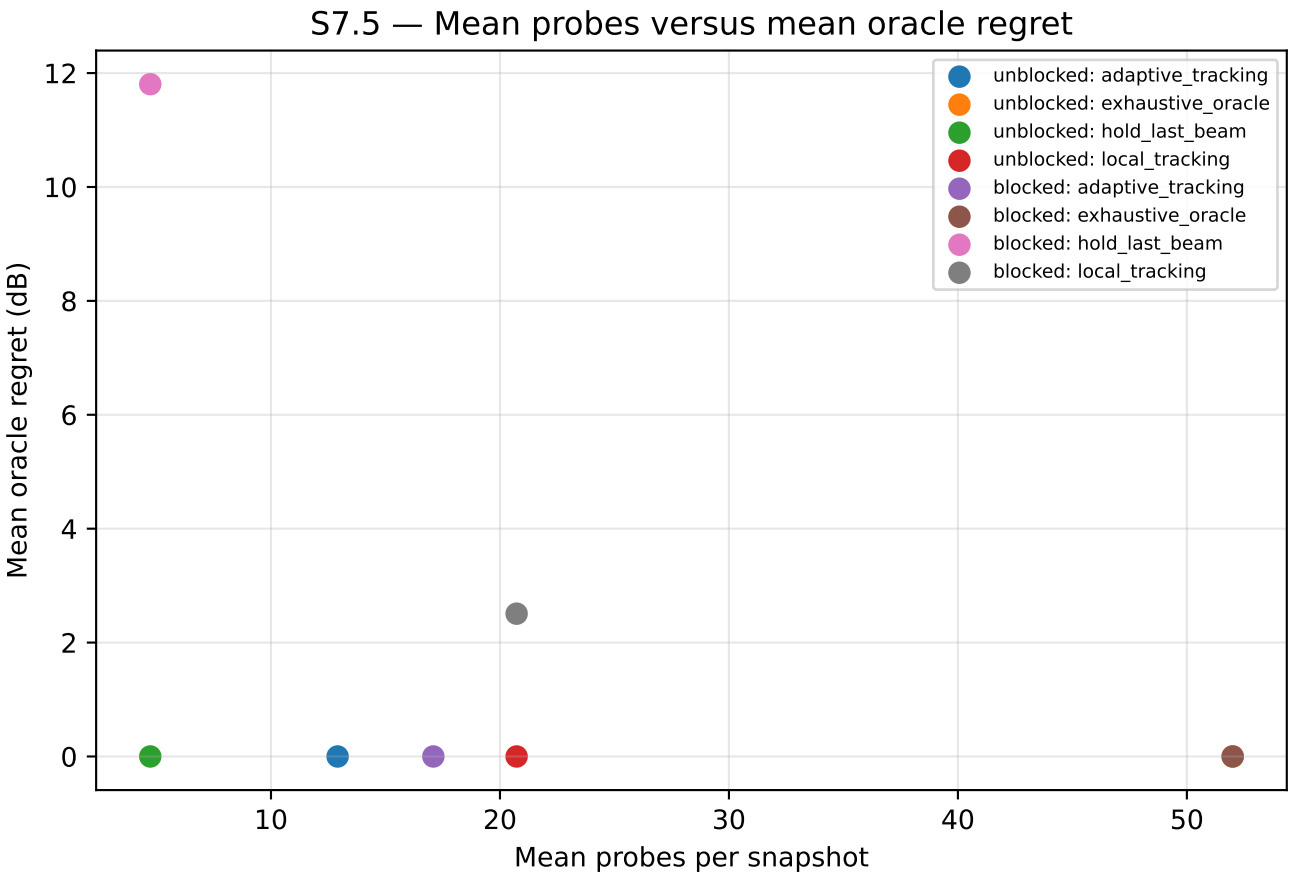

S7.2 — Best codebook beam gain by receiver position
unblocked
blocked
−80
−90
−100
−110
−120
−130
10 log10(|hᵀw|²) (dB of dimensionless gain)
17.5
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5
Receiver X position (m)

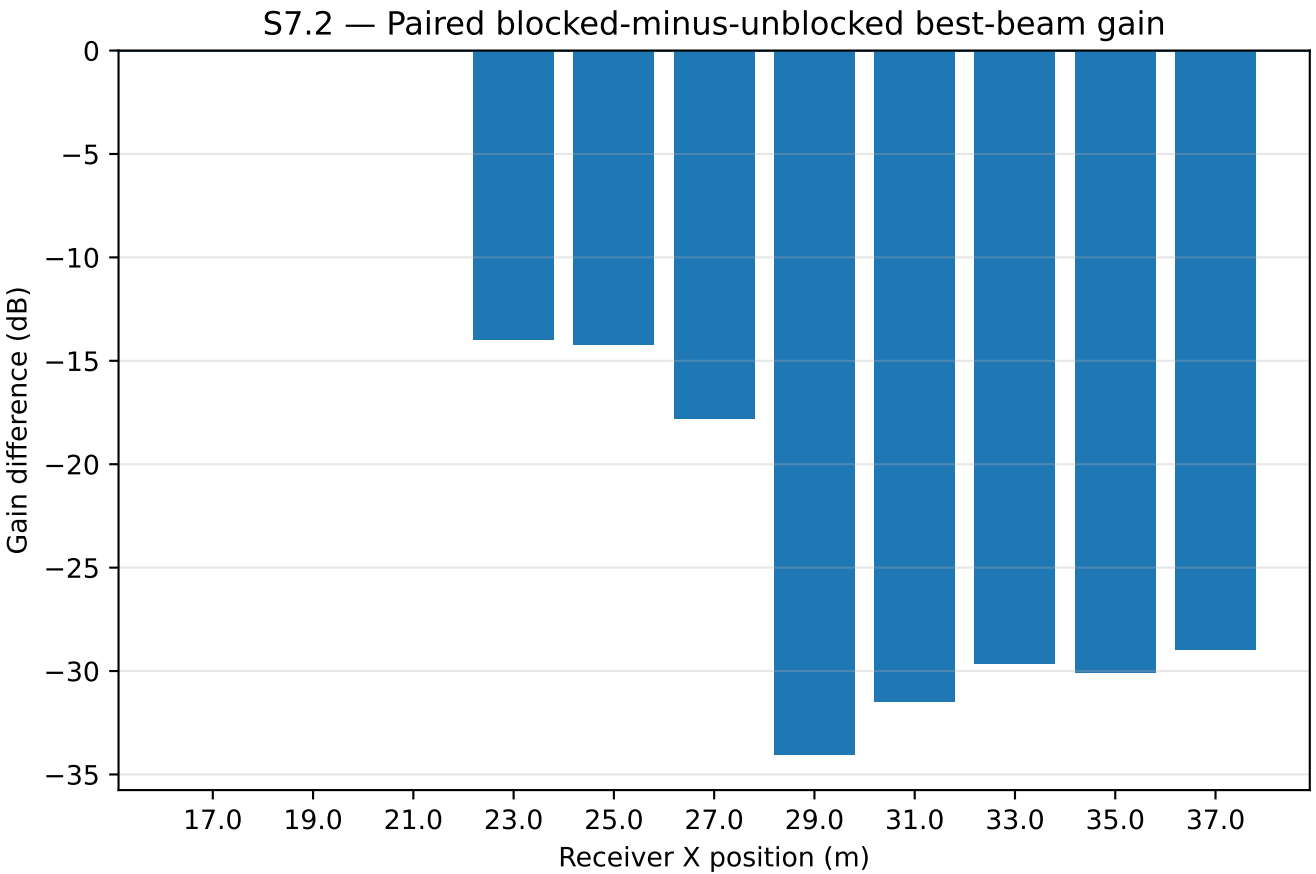

S7.1 — Power-weighted RMS path-delay spread
unblocked
blocked
RMS delay spread (ns)
0
20
40
60
80
100
17.5
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5
Receiver X position (m)


S7.3 — Ray interaction event counts
condition
blocked
unblocked
Interaction events
0
50
100
150
200
250
300
refraction
specular_reflection
Interaction type


S7.4 — Beam-selection regret
unblocked
blocked
adaptive_tracking
hold_last_beam
local_tracking
Oracle regret (dB)
Receiver X position (m)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
17.5
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5


S7.5 — Beam-training measurement count
unblocked
blocked
Beam probes per snapshot
Receiver X position (m)
0
10
20
30
40
50
17.5
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5
adaptive_tracking
exhaustive_oracle
hold_last_beam
local_tracking


S7.5 — Mean probes versus mean oracle regret
Mean oracle regret (dB)
Mean probes per snapshot
0
2
4
6
8
10
12
10
20
30
40
50
unblocked: adaptive_tracking
unblocked: exhaustive_oracle
unblocked: hold_last_beam
unblocked: local_tracking
blocked: adaptive_tracking
blocked: exhaustive_oracle
blocked: hold_last_beam
blocked: local_tracking


S7.2 — Paired blocked-minus-unblocked best-beam gain
Gain difference (dB)
0
−5
−10
−15
−20
−25
−30
−35
17.0
19.0
21.0
23.0
25.0
27.0
29.0
31.0
33.0
35.0
37.0
Receiver X position (m)